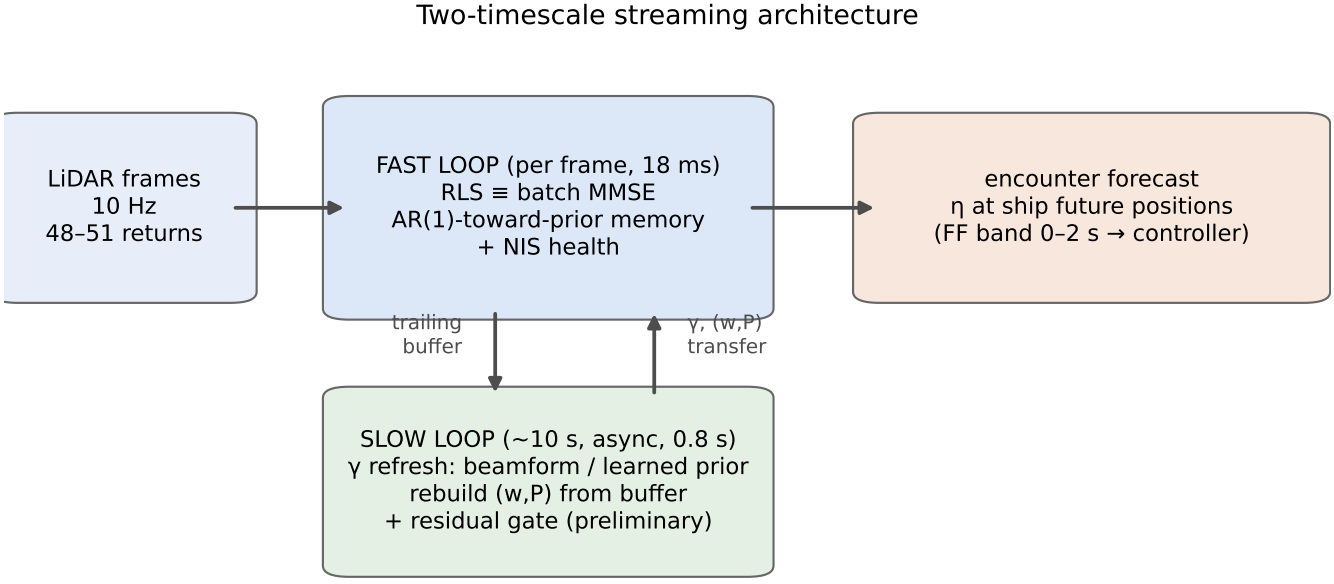
Two-timescale streaming architecture
LiDAR frames
10 Hz
48–51 returns
FAST LOOP (per frame, 18 ms)
RLS ≡ batch MMSE
AR(1)-toward-prior memory
+ NIS health
encounter forecast
η at ship future positions
(FF band 0–2 s → controller)
trailing
buffer
γ, (w,P)
transfer
SLOW LOOP (~10 s, async, 0.8 s)
γ refresh: beamform / learned prior
rebuild (w,P) from buffer
+ residual gate (preliminary)